

THE HILLS GRAMMAR SCHOOL



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16

THE HILLS GRAMMAR SCHOOL

TRIAL HSC
1999

MATHEMATICS

3 UNIT

TIME ALLOWED: 2 Hours (plus 5 minutes reading time)

Teacher Responsible: Mrs B Spencer
Mrs S Maxton

INSTRUCTIONS:

- Attempt all questions.
- In every question show all necessary working.
- Silent calculators and approved templates may be used.
- Start each question on a NEW page and hand up your paper in ONE bundle with your name marked on EVERY page.
- All questions are of equal value.

Question 1**Marks**

- a) Express $\frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}} + \frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}}$ in the form of $a\sqrt{5} + b\sqrt{6}$ 2
- b) Show that $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$ 2
- c) Find the acute angle between the lines
 $y = 4x - 2$ and $2x + 3y - 9 = 0$
to the nearest minute. 2
- d) Find the co-ordinates of the point P which divides the interval AB externally in the ratio 3:1. A is (-4,2) and B is (6,5). 2
- e) (i) What is the maximum value of $5\sin\theta + 12\cos\theta$; and 2
(ii) What is the first positive value of θ for which this maximum occurs. 2

Question 2

- a) Find $\int x^2 \sqrt{(x^3 - 9)} dx$ using $u = x^3 - 9$ 3
- b) Evaluate $\int_0^{\frac{\pi}{12}} \cos^2 x \, dx$. Answer in exact value terms. 3
- c) Find the term independent of x in the expansion of $(x^2 - \frac{1}{x})^{12}$. 3
- d) The remainder when $x^3 + ax + b$ is divided by $(x-2)(x+3)$ is $2x+1$. Find a and b . 3

Question 3

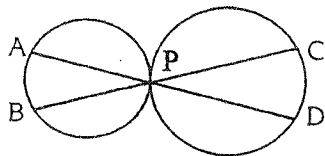
Marks

- a) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The angle POQ is 90° . O is $(0,0)$. Show that $pq = -4$ and hence find the equation of the locus of M , the midpoint of PQ . 2
3
- b) A projectile is fired from O with a velocity of 20m/s at an angle of 60° . The projectile just clears a wall 25m from the point of projection. How high is the wall? Take $g = 10\text{m/sec}^2$. 3
- c) $f(x) = x^3 - x^2 - x - 1$.
- (i) Show that the equation $f(x) = 0$ has a root in the interval $1 < x < 2$. 1
- (ii) Use Newton's method once to find a better approximation for the root, taking x to be 1.5 . 3

Question 4

- a) A stone is thrown into a pond and creates a circular ripple which expands so that $\frac{dr}{dt} = 1.5\text{m/s}$. Find the rate at which the area of the circle is increasing when the radius is 2m . Answer to 4 significant figures. 2
- b) A particle moving along the x -axis has a velocity given by $v^2 = 15 - 2x - x^2$.
- (i) Show that the centre of the motion is at $x = -1$. 2
- (ii) Show that $a = -n^2 x$ and thus the motion is simple harmonic. 2
- (iii) Find the amplitude and period of the motion. 2

c)



Two unequal circles touch at P . APD and BPC are straight lines.

- (i) Copy the diagram in a larger scale. 4
- (ii) Draw the common tangent XPY .
- (iii) Prove that AB is parallel to CD .

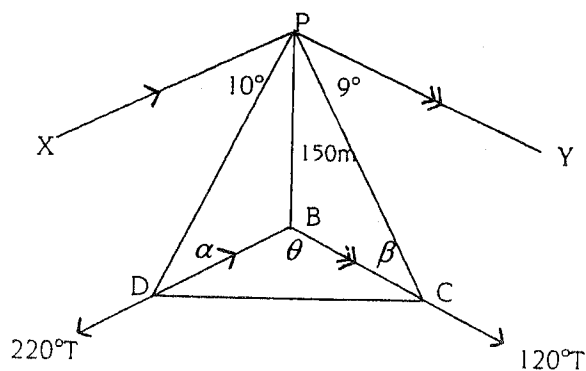
Question 5

Marks

- a) Use the expansion $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ to prove that
 $1 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n = 0$. 2
- b) In how many ways can a committee of 4 people be selected from a group of 10 people if;
- 2 particular members P and Q are included; 3
 - 2 particular members P and Q are excluded. 3
- c) I participate in a raffle where my chances of winning a prize are one in ten. If I buy a book of 12 tickets, what is the probability of my winning exactly two prizes. Leave answer in index form. 4

Question 6

a)



A ship is observed from the top of a 150m cliff BP with an angle of depression 9° when the ship is at the point C. Ten minutes later it is seen at D with an angle of depression of 10° . $\angle PDB = \alpha$, $\angle PCB = \beta$ and $\angle DBC = \theta$. BC bears 120°T . BD bears 220°T .

- Show that $\alpha = 10^\circ$, $\beta = 9^\circ$ and $\theta = 100^\circ$. 3
 - Show that $BD = 150 \cot \alpha$ and $BC = 150 \cot \beta$ and hence that
 $CD^2 = 150^2 (\cot^2 \alpha + \cot^2 \beta - 2 \cot \alpha \cot \beta \cos \theta)$. 4
 - Find the speed of the ship in km/h to three significant figures. 2
- c) Solve for all x , $2\sin^2 x = \sin 2x$. Answer in radian measure. 3

Question 7

Marks

- a) My loungeroom is kept at a constant temperature of 25°C . A cup of tea left standing in the room cools at a rate proportional to the difference in temperature between the tea and its surroundings so that
- $$\frac{dT}{dt} = k(T - 25).$$
- After 20 minutes the temperature of the tea has dropped from 95°C to 65°C .
- (i) Show that $T = 25 + Ae^{kt}$ is a solution of $\frac{dT}{dt} = k(T - 25)$. 1
- (ii) Find the values of A and k . 3
- (iii) Find the temperature of the tea after a further 10 minutes. 2
- b) Consider the function $f(x) = x \sin^{-1} x$.
- (i) Show that $f(x)$ is an even function. 1
- (ii) Find $f'(x)$ and hence the co-ordinates of the only turning point. 2
- (iii) Determine the domain and range of $f(x)$. 2
- (iv) Sketch $f(x)$. 1

Solution. Hills Grammar

Question 1

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Trial HSC.

a) $\frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}} + \frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}}$

$$= \frac{\sqrt{5}-\sqrt{3}+\sqrt{10}-\sqrt{6} + \sqrt{5}+\sqrt{3}-\sqrt{10}-\sqrt{6}}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})}$$

$$= \frac{2\sqrt{5}-2\sqrt{6}}{2}$$

$$= \sqrt{5}-\sqrt{6} \quad a=1 \quad b=-1$$

(2)

b) $\frac{nCr}{nCr-1} = \frac{n!}{(n-r)!r!} \div \frac{n!}{(n-r+1)!(r-1)!}$

$$= \frac{n!}{(n-r)!r!} \times \frac{(n-r+1)!(r-1)!}{n!}$$

$$= \frac{n-r+1}{r} \text{ or } \frac{n-r+1}{r}$$

(2)

c) $y=4x-2$
 $2x+3y=9$

$$m_1 = \frac{4}{1}$$

$$m_2 = -\frac{2}{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{4 + \frac{2}{3}}{1 - \frac{8}{3}} \right|$$

$$= \frac{14}{5}$$

$$\theta = 70^\circ 21'$$

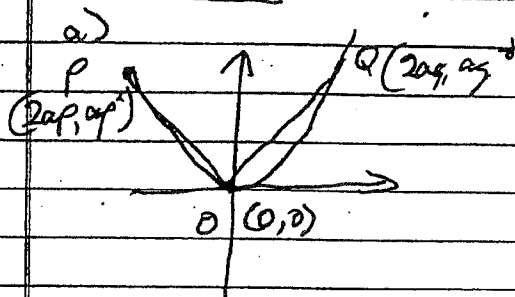
(2)

d) $A(-4, 2) \quad B(6, 5) \quad P(15, 6)$

(2)

e) $5 \sin \theta + 12 \cos \theta = 13 \left(\frac{5}{13} \sin \theta + \frac{12}{13} \cos \theta \right)$
 $= 13 \sin(\theta + \alpha)$ where $\alpha = 67^\circ 23'$

Question 3



$$PO \perp QO$$

$$\therefore m_1 \times m_2 = -1$$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$pq = -4$$

(2)

M is $a(p+q)$, $\frac{a(p^2+q^2)}{2}$

$$x = a(p+q) \quad 2y = a(p^2+q^2)$$

$$\frac{x^2}{a^2} = p^2 + q^2 = 8$$

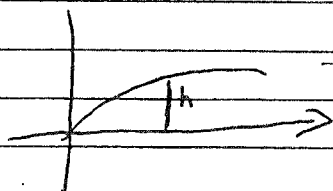
$$\frac{x^2}{a^2} + 8 = \frac{2y}{a}$$

$$x^2 = 2ay - 8a^2$$

$$x^2 = 2a(y - 4a)$$

(3)

b)



$$x = vt \cos \theta$$

$$25 = 20t \times \cos 60^\circ$$

$$t = 2.5$$

$$h = -5t^2 + 20 \times 2.5 \sin 60^\circ$$

$$h = 12.05 \text{ m}$$

(3)

c) (i) $f(x) = x^3 - x^2 - x - 1$

$$f(1) = -2 \quad \text{negative}$$

$$f(2) = 1 \quad \text{positive}$$

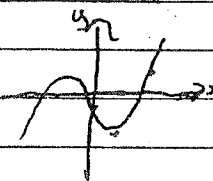
$\therefore$ there is a root between 1 and 2

(1)

(ii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$= 1.5 + \frac{1.375}{2.75}$$

$$= 1.5$$



$$f(x) = x^3 - x^2 - x - 1$$

$$f(1.5) = -1.375$$

$$f'(x) = 3x^2 - 2x - 1$$

$$f'(1.5) = 2.75$$

(3)

Question 5.

a) $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

Let $x = -1$

(2)

$$0 = 1 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n \text{ as reqd.}$$

b) (i) If A and B are included, I need only choose 2 from the other eight

$${}^8C_2 = 28 \text{ ways}$$

(3)

(ii) If A and B are excluded, I have 8 choose 4 from the remaining 8

$${}^8C_4 = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1}$$

$$= 70 \text{ ways}$$

(3)

2) $p = \text{win } \frac{1}{10}$

$q = \text{lose } \frac{9}{10}$

$(p+q)^{12}$ select the p^2 term.

$${}^{12}C_{10} p^2 q^{10}$$

(14)

$$= \frac{12 \times 11}{2 \times 1} \times \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^{10}$$

$$= \frac{66 \times 9^{10}}{10^{12}}$$

Question 6

(i) $\angle P D B = \angle P B A$ (alternate $\angle$ s $\because P D \parallel P B$)

(1)

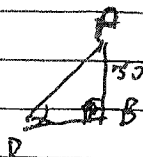
$\angle P C B = \angle P B A$ (" $\because P Y \parallel B A$)

(1)

$\angle D B A = 220^\circ - 120^\circ = 100^\circ$ (given bearing)

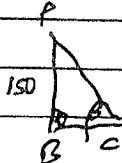
(1)

(ii)



$$\cot \alpha = \frac{DB}{150}$$

$$DB = 150 \cot \alpha$$



$$\cot \beta = \frac{BC}{150}$$

$$BC = 150 \cot \beta$$

(2)

By cosine rule

By cosine rule

$$CD^2 = DB^2 + BC^2 - 2 DB BC \cos \theta$$

Question 7

a) $\frac{dT}{dt} = k(T-25)$

(i) $T = 25 + Ae^{kt}$ $Ae^{kt} = T-25$
 $\frac{dT}{dt} = kAe^{kt}$
 $= k(T-25)$ as reqd. (1)

(ii) When $t=0$ $T=95$
 $95 = 25 + A$ $A=70 \dots (1)$
 when $t=20$ $T=65$
 $65 = 25 + 70e^{20k}$
 $\frac{40}{70} = e^{20k}$
 $k = \frac{\ln \frac{4}{7}}{20}$ ~~0.02798~~ R
 $= 0.02798$

(iii) When $t=30$
 $T = 25 + 70e^{30k}$
 $= 55$ (2)

b) $f(x) = x \sin^{-1} x$

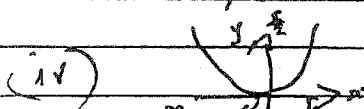
(i) $f(x) = x \sin^{-1} x$
 $f(-x) = -x \sin^{-1}(-x)$
 $= -x x \sin^{-1}(x)$
 $= x \sin^{-1} x$
 $\therefore$ function is even. (2)

(iii) $f'(x) = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$
 $f'(x) = 0$ when $x=0$
 $\therefore$ F.P. @ $(0,0)$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y	-1.1	0	1.1
shape	1	-	1

Min T.P.

(iii) domain $-1 \leq x \leq 1$
 range $0 \leq y \leq \frac{\pi}{2}$ (2)



(1)